



Installation, Operating and Maintenance
Instructions
MaxFire™ Multi-Fuel Room Heater



WARNING! Please read this entire manual before installation and use of the Bixby MaxFire™ Multi-Fuel Room Heater. Failure to follow these instructions could result in property damage, bodily injury or even death. Contact local building officials, fire officials or the authority having jurisdiction about restrictions, permits and installation inspection requirements in your area.

Save these instructions for future reference.

PLEASE RECORD THE FOLLOWING INFORMATION:

MODEL _____ DEALERSHIP _____ INSTALLER _____
DATE PURCHASED _____ DEALER PHONE _____ INSTALLER PHONE _____
SERIAL NUMBER _____

Bixby Energy Systems
877-500-2800
www.bixbyenergy.com

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SAFETY INFORMATION

1. Install and use only in accordance with these installation and operating instructions, page 9. Contact local building officials, fire officials, or the authority having jurisdiction about restrictions, permits, and inspection requirements in your area.
2. Refer to instructions and local codes for precautions required for passing the vent pipe through a combustible wall or ceiling. Inspect and clean vent system frequently in accordance with manufacture's instructions.
3. DO NOT INSTALL A FLUE DAMPER IN THE EXHAUST VENTING SYSTEM OF THIS UNIT.
4. DO NOT CONNECT THIS UNIT TO A CHIMNEY SERVING ANOTHER APPLIANCE.
5. INSTALL VENT AT CLEARANCES SPECIFIED IN THE VENTING SECTION.
6. For use with only whole corn kernels, wood pellet fuel or approved pelletized biomass.
7. DO NOT BURN GARBAGE, OR FLAMMABLE FLUIDS SUCH AS GASOLINE, NAPTHA OR ENGINE OIL IN YOUR BIXBY ROOM HEATER. Store any fuel away from your heater.
 - Pellet stove starter gel may be used to assist in the lighting of the Room Heater. See lighting instructions.
8. Always unplug the heater before cleaning or servicing.
9. Dispose of all ashes in a metal container.
10. The use of a protective screen (i.e. fireplace screen) is recommended when children or animals are present.
11. DO NOT INSTALL IN A SLEEPING ROOM.
12. The door must be closed and latched during operation. Always allow the Room Heater to cool before opening the door to clean the firebox and / or the glass. If door must be opened during operation, use caution as embers may escape from combustion chamber.

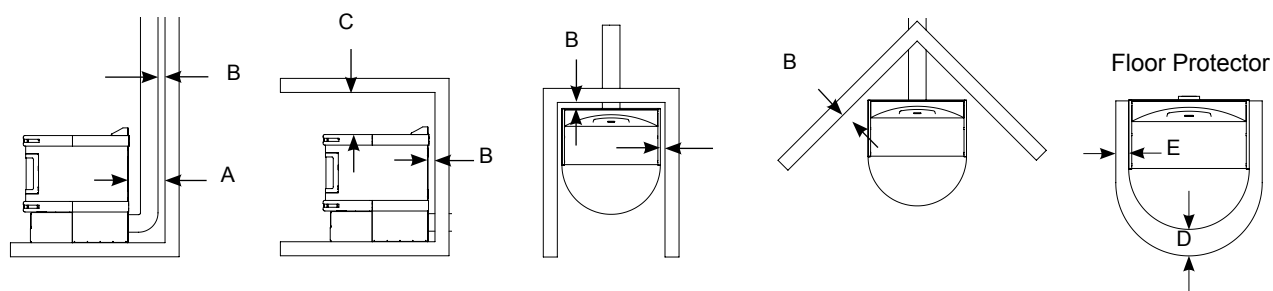
WARNING: Biomass (i.e. corn, wood) is a solid material and by its very nature will accumulate ash that must be cleaned out of the Room Heater and venting on a regular basis. See cleaning and maintenance schedule. FAILURE TO DO SO MAY VOID WARRANTY.

NOTE: It is recommended that a carbon monoxide detector be installed to ensure protection from the hazards of carbon monoxide. It is also recommended that a smoke detector be installed for early warning in case of fire.



FRONT SURFACES OF THE BIXBY ROOM HEATER WILL BE HOT WHEN THE UNIT IS IN OPERATION. DO NOT TOUCH, AND KEEP CHILDREN AND CLOTHING AWAY. CONTACT MAY CAUSE SEVERE BURNS. KEEP FURNISHINGS AND OTHER COMBUSTIBLE MATERIALS AT LEAST 2 FT. (610 MM) DISTANCE AWAY FROM THE BIXBY ROOM HEATER.

MINIMUM CLEARANCES TO COMBUSTIBLE MATERIALS



Free Standing and Alcove (rear only) Installation Clearances:

A	10.5"	(267mm)	D	6"	(150mm)
B	2"	(50mm)	E	3"	(75mm)
C	12"	(305mm)			

NOTE: Minimum clearances are for temperature concerns and do not allow for enough room for maintenance. Please allow enough room on the sides for routine maintenance, 6" - 12" (150mm - 305mm) of clearance to non-movable objects (i.e. walls) is recommended. Also, clearance to the back of the Room Heater is needed for replacement of the filter.

Use a non-combustible floor protector under the unit, extending 3" (76mm) past the side of unit (or less if installed at closer clearances) and 6" (150mm) in front of Room Heater door.

Alcove Dimensions (Rear Vent Only):

Minimum Height - 42" (1065mm), Minimum Width - 32" (815mm),
Maximum Depth - 24" (610mm)



SPECIFICATIONS

Fuel: Multi-fuel System: Dry shelled corn, Wood Pellets (recommended maximum size approximately ¼ in. diameter x 1¼ in. long), or “BIXBY” certified Blue or Green Biomass pellets. The use of “cracked corn” is not recommended due to feed and burn issues. The use of lower quality corn with foreign material (cobs, stones, or other material) may cause feed issues. In addition, the use of seed corn is not recommended due to the coating interfering with the operation of the Room Heater. When loading the corn into the hopper, remove as much foreign material as possible. See Operating Instructions (page 21) for more information.

Level	Approx. Feed Rate lbs/hr (kg/hr)	Time Between Ash Dumps *	Approximate Hours/ Full Load *
1	1.8 (0.8)	21 hrs	57
2	2.1 (1.0)	17 1/2 hrs	48
3	2.3 (1.1)	16 hrs	43
4	2.6 (1.2)	14 hrs	38
5	3.0 (1.4)	12 1/2 hrs	34
6	3.5 (1.6)	10 1/2 hrs	29
7	3.9 (1.8)	9 1/2 hrs	26
8	4.6 (2.1)	8 hrs	22

*Assumes running at a single level, actual time will vary as the burn level changes, starting and shutting down, and trim knob adjustments.

Heating Area: Varies depending on floor plan, heat loss, geographic location, altitude, heater venting configuration, moisture content of fuel and climate.

Installation: Install in accordance with “BIXBY” instructions and local building codes.

Shipping Weight: 380 pounds (172 kg)

Electrical: 120 VAC, 60 Hz., 15 amp Service Required, dedicated line recommended
Start: 10 Amps
Run: 2 Amps

Dimensions: Height = 33 in (838 mm)
Width = 28 in (711 mm)
Depth = 30 in (762 mm)

Fuel Hopper Capacity: 2.4 cu. ft. (approx. 106 lbs. shelled corn)

Listing: The Bixby MaxFire™ Multi-Fuel Room Heater was tested by OMNI-Test Laboratories, Beaverton, Oregon to ASTM E1509 and ULC C1482 (report number 223-S-06-2). Emission tests were also conducted per the USA Environmental Protection test methods 28A and 5G for particulate emissions. The Max-Fire™ is exempt under 40 CFR 60.531, therefore the results have not been submitted to the EPA.

The results are as follows:

Category 1 (low burn)	2.3 grams per hour (.081 ounces)
Category 2 (medium low burn)	2.0 grams per hour (.070 ounces)
Category 3 (medium high burn)	2.42 grams per hour (.085 ounces)
Category 4 (high burn)	2.21 grams per hour (.077 ounces)
Overall weighted average	2.25 grams per hour (.079 ounces)

IMPORTANT:

Proper material handling equipment must be available to safely transport and position the Bixby Room Heater.

1. Remove the protective packaging, using appropriate tools, and detach the unit from its shipping pallet. The Room Heater is attached to the pallet with two bolts. Remove the side pannels to gain access to the bolts.
2. Using a suitable transporting device, move the unit to the desired location, centering it on the Bixby Hearth Pad or an alternate protective plate. Care should be taken not to mar the Hearth Pad. Placing protective material under the Bixby Room Heater when positioning is recommended.
3. Use the included bulls-eye level to level the unit by adjusting the height of its legs. Leveling the unit is required to maintain even fuel distribution. See Page 8 for more information.

MOBILE HOME INSTALLATION

WARNING: DO NOT INSTALL IN A SLEEPING ROOM.

CAUTION: THE STRUCTURAL INTEGRITY OF THE MOBILE HOME FLOOR, WALLS AND CEILING / ROOF MUST BE MAINTAINED (i.e. Do not cut through floor joists, wall studs, ceiling trusses, etc.)

1. A Bixby Direct-Vent system must be used to provide an outside air inlet for combustion air and must be unrestricted.
2. The Room Heater must be secured to the mobile home by bolting it to the floor (using lag bolts).
3. The Room Heater must be grounded with #8 copper grounding wire or equivalent, terminated at each end with an NEC approved grounding device.
4. Refer to pages 16 and 17 for typical horizontal or vertical mobile home venting installation.
5. Refer to clearances to combustibles section on pages 3 and 11 for distances to combustibles and appropriate chimney systems.
6. Seal all wall and floor inlets to prevent air or moisture penetration. Check periodically to ensure inlet is free of obstruction, such as snow and ice.
7. Installation should be in accordance with the Manufactured Home and Safety Standard (HUD), CFR 3280, Part 24.

PLANNING THE INSTALLATION

The Bixby Max-Fire™ Room Heater is intended for use in buildings, manufactured homes or for mobile home installation (see previous page for mobile home installation instructions).

CAUTION: The Max-Fire™ Room Heater should not be installed at a location where it could come in contact with curtains, drapes, walls, carpeting or other combustible surfaces. The Room Heater must not be installed in a sleeping room. When choosing a location, proximity to an electrical outlet (do not use an extension cord) and the best location for the venting outlet and its connecting piping should be considered. See Venting Installation and Instructions, page 9.

CAUTION: This Room Heater must be installed only by an authorized dealer, and/or by individuals who are technically qualified and versed in any local codes or regulations that may apply to installation and application of heating equipment of this type. Clearances specified in this manual are minimum and any reduction must be approved by the regulatory authority.

To assure safe operation, it is absolutely essential that a floor protective device of metal or other non-combustible material be in place beneath the Room Heater. Note drawing below to determine dimensions the size of the mat. Lightweight, properly sized hearth pads are available from Bixby Energy Systems in colors to match your Room Heater.

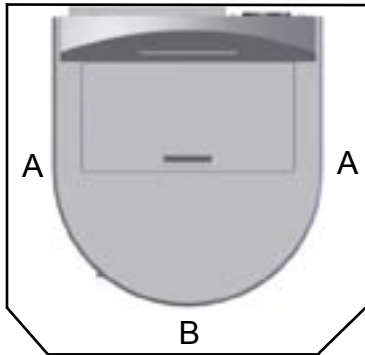


Figure 1

The non-combustible floor protector should extend under unit 3 in. (76mm) to each side of unit when installed in a free standing location and 6 in. (152 mm) in front of Room Heater door. The floor protector can be less than 3 in.(76mm) if installed in an alcove. See Minimum Clearances, page 3.

A = 3 in. (76mm)

B = 6 in. (152mm)

PERFORMING THE INSTALLATION

The unit is shipped completely assembled. It is enclosed in protective packaging material and attached to a shipping pallet. A shop cart or other suitable material handling device (minimum capacity of 400 lbs. (180kg)) will be needed to transport the Bixby Room Heater to its desired location. To bring the Room Heater into the room chosen, a door opening of at least 2 ft. 8 in.(80cm) will be needed. Appropriate tools will be needed to connect the venting pipes and fittings, as well as a hole-cutting device for venting the piping to the outside. The side of the shipping box can be used to help locate the Room Heater and the vent hole location.

OPTIONAL THERMOSTAT OPERATION

The Room Heater is designed so that it can be operated by a wall mounted thermostat. This option is not provided directly from the factory, but can be purchased from your dealer or the Bixby website (www.bixbyenergy.com) or any other “on/off” 24V AC wall mounted thermostat can be used. **Do not use a powered thermostat to avoid damage to the Room Heater.** Follow the the thermostat installation instructions carefully.

NOTE: Thermostat should be mounted on an inside wall and not in direct line with the Room Heater convection air. A preferred location would be on an inside wall opposite the Room Heater.

When the thermostat is hooked up and calling for heat, the Room Heater will display the currently selected level using the LEDs. If the thermostat is NOT calling for heat, the currently selected level will be displayed, but all the LEDs will be slowly flashing in unison and the Room Heater run at level 1.

THERMOSTAT INSTALATION INSTRUCTIONS

1. Unplug the Room Heater.
2. Remove the left side panel.
3. Locate the Trim Pot / Thermostat bracket.
4. Loosen the middle two screws on the thermostat block and remove the metal jumper.
5. Run the wire for the thermostat through the black grommet on the back panel.
6. Attach the wires of the thermostat to the two terminals.
7. Replace the side panel.

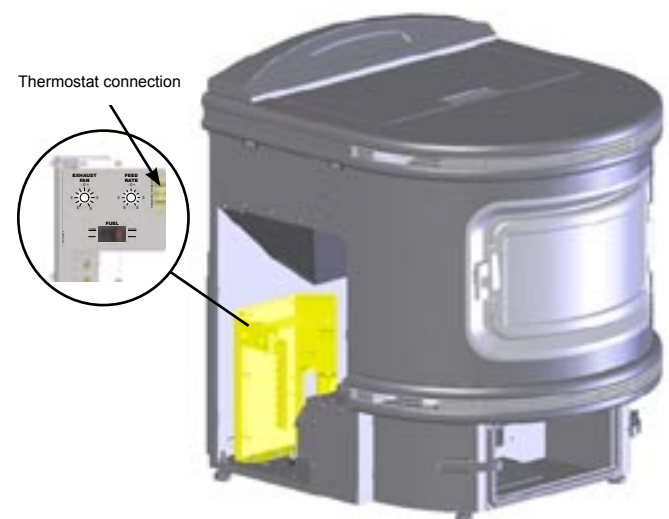


Figure 2

LEVELING THE ROOM HEATER

CAUTION: THE ROOM HEATER SHOULD BE LEVELED BEFORE ATTACHING THE VENT PIPE.

The supplied bulls eye level can be used to level the Room Heater front-to-back and side-to-side, a 6 in. level may also be used.

1. Place the level centered side-to-side and about 4 in. (10cm) back from the front of the Room Heater on the top surface. The bubble should end up centered inside the middle circle on the level. If the bubble is not in that position, the bubble will be towards the high side of the Room Heater. See Figure 3.
2. Locate the leveling feet on the bottom of the Room Heater. See Figure 4.
3. Lift the Room Heater near the foot to be raised (low side, opposite the bubble on the level). A pry bar can be used to lift the Room Heater with a wooden shim or something similar to protect the hearth pad.
4. With the corner of the Room Heater elevated, place a piece of wood (or other blocking) under the Room Heater to hold it while adjusting the leg. The leg length should be adjusted by hand, but if needed a wrench maybe used. This will not work if the foot is turned all the way in, as that portion will not be accessible.
5. Remove the block using the prybar and check adjustment.
6. Proceed to adjust the feet until the bubble is centered inside the inner circle. This is a very important step as the feed system is gravity fed and it will ensure that the fuel is fed into the burn pot correctly.

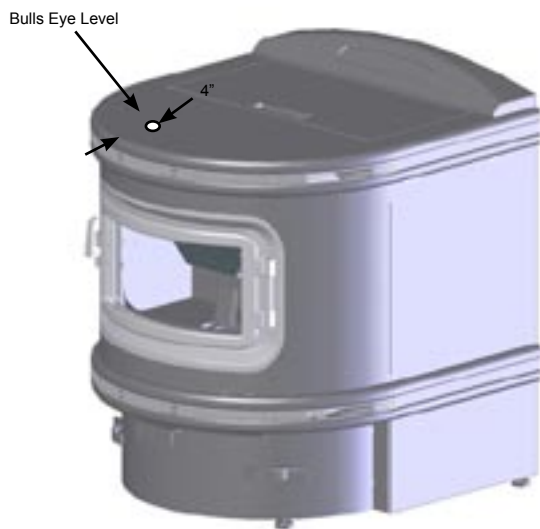


Figure 3

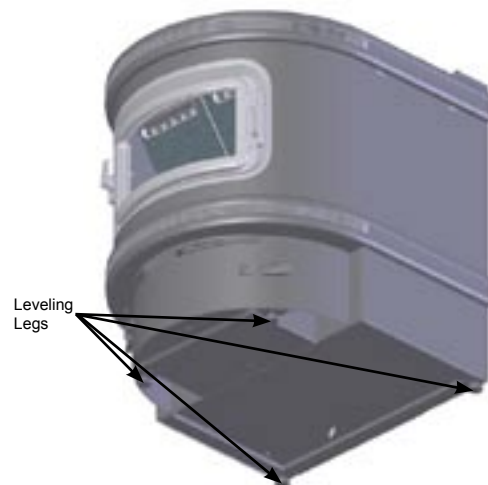


Figure 4

VENTING INSTALLATION AND INSTRUCTIONS

WARNING:

A MAJOR CAUSE OF VENT RELATED FIRES IS FAILURE TO MAINTAIN REQUIRED CLEARANCES (AIR SPACES) TO COMBUSTIBLE MATERIALS. IT IS OF THE UTMOST IMPORTANCE THAT THIS VENT SYSTEM BE INSTALLED ONLY IN ACCORDANCE WITH THESE INSTRUCTIONS.

VENT SYSTEM APPROVALS

BIXBY DIRECT VENT PIPE is listed by OMNI-Test Laboratories, Inc. as a vent for BIXBY pellet and corn Room Heaters only. NO OTHER PIPE IS LISTED FOR USE WITH BIXBY ROOM HEATERS. The use of non-Bixby vent pipe will void the Room Heater's warranty. The minimum clearance from this vent to combustible materials is 2 in. (50mm). Combustible materials include but are not limited to lumber, plywood, sheetrock, plaster and lath, furniture, curtains, electrical wiring, and building insulation of any kind. Never fill any required clearance space with insulation or any other materials.

This vent is tested and listed by OMNI-Test Laboratories, Inc. to UL Standard 641 and to ULC/ORD-C441 (Canada) for safe low temperature venting systems, Type L, and produced under the factory inspection and follow-up program of OMNI-Test Laboratories, Inc.

NEVER INSTALL SINGLE-WALL PIPE TO THE Room Heater.

NOTE:

Proper planning for your vent installation will result in greater safety, efficiency, and convenience, saving both time and money. Use only authorized BIXBY ENERGY SYSTEMS, INC. listed parts. Do not install damaged parts.

TOOLS AND EQUIPMENT

Level (a six inch level is recommended)
Eye protection
Hammer
Screwdriver
Tape measure
High temperature waterproof sealant (min 500F)
Saber or keyhole saw
Stud finder
Adjustable wrench
Gloves

CAUTION: The Room Heater should be leveled before attaching the vent pipe. See page 8 for leveling information.

VENTING COMPONENTS

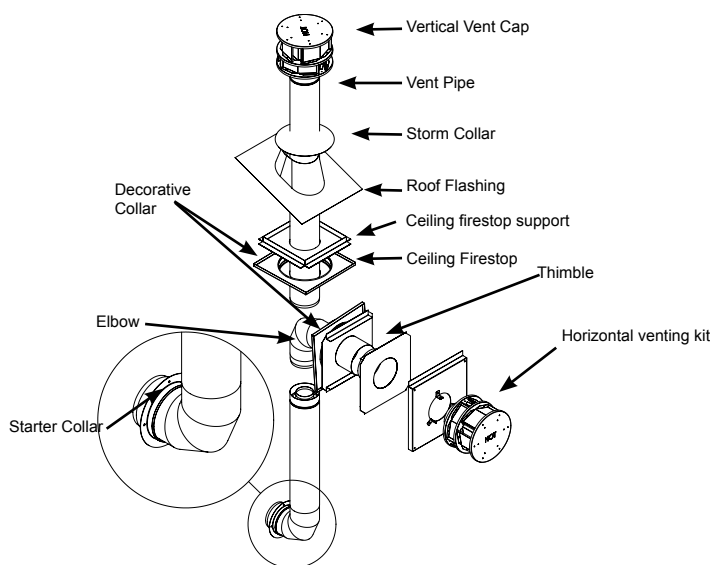


Figure 5

INSPECTIONS

The use of pelletized fuel does not eliminate the need for inspection and cleaning. During the heating season, inspect monthly.

PERMITS

Contact your local building officials, fire officials, or the authority having jurisdiction about restrictions, permits, and inspection requirements in your area.

VENT MAINTENANCE

- **It is essential to have your venting and vent cap cleaned every year (each Spring) to remove soot build up.** If you have doubts about your ability to clean it, contact a professional chimney sweep. Use a plastic, wood, or flexible steel brush. Do not use a stiff brush that will scratch the stainless steel liner of your system.
- Bixby Direct Vent systems must be installed so that access is provided for inspection and cleaning.
- The vent system should be inspected at least once every month during the heating season.
- Chemical cleaners must be used with great caution. Use only those which are absolutely guaranteed not to corrode or have any other harmful effects.
- In case of fire, shut off appliance and call your Fire Department. Do not use the appliance or vent until it has been inspected for possible damage.
- Due to the nature of solid fuels and their production of ash, Bixby Energy Systems is not responsible for flue by-products that might discolor roofs or walls.

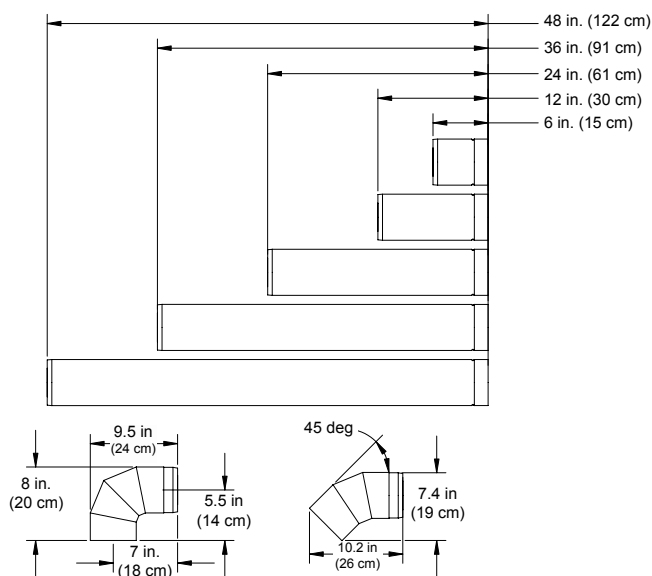


Figure 6

VENT LOCATION

When exiting through walls, make sure NFPA rules are followed for distance from windows and openings.

NOTE:

The termination cap will be HOT during operation. Consider it's proximity to doors or other traffic areas.

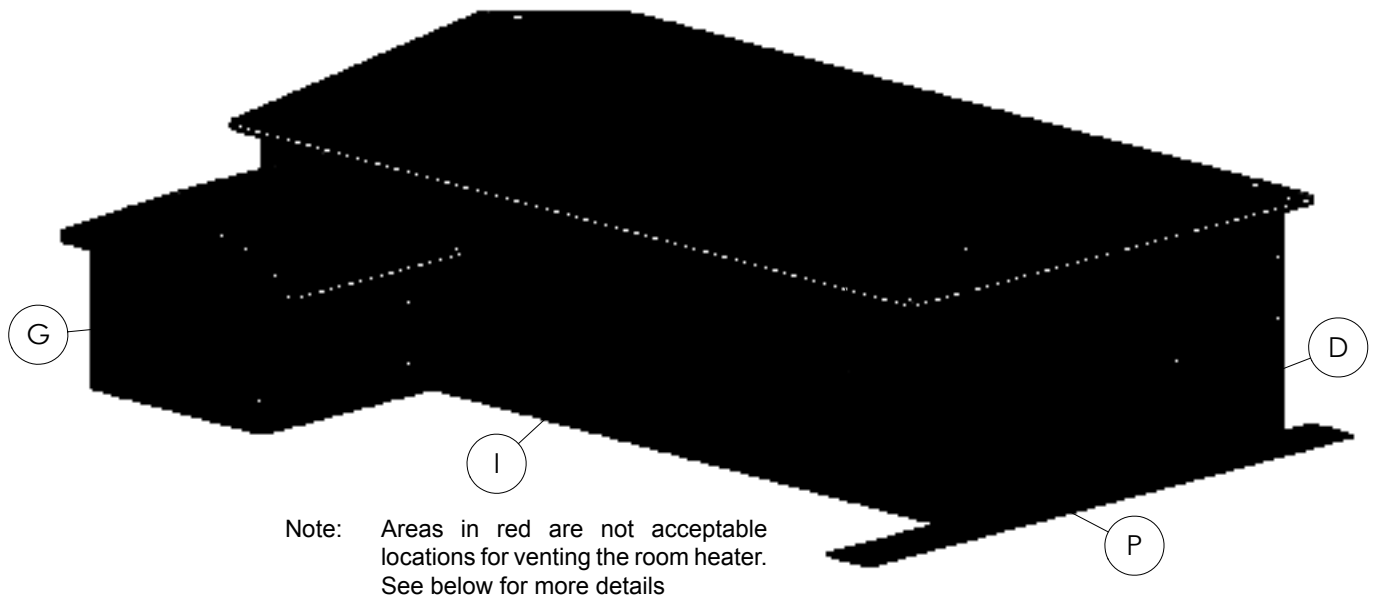


Figure 7

Follow the below listed NFPA rule for distance of exit terminal from windows and openings:

Per NFPA 211 10.4.1 all vents shall terminate above the roof surface (vertical venting). However, per NFPA 211 10.4.5, Mechanical draft systems shall not be required to comply with 10.4.1 provided they comply with the following:

To ensure compliance with NFPA 211, the vent, shall be located in accordance with the following:

- Not less than 3 ft. (.91m) above any forced air inlet (**I**) located within 10 ft. (3m).
- Not less than 2 ft. (.61m) above grade, non combustibile material ground cover is recommended below the vent if less than 3 ft. (.91m) above grade.
- Not less than 4 ft. (1.2m) below, 4 ft. (1.2m) horizontally from or 1 ft. (.3m) above any door (**D**), window (**W**) or gravity air inlet into any building.
- Not less than 2 ft. (.61m) from an adjacent building and not less than 7 ft. (2.1m) above grade when located adjacent to public walkways / paths (**P**).
- Not within 6 ft. (1.8m) of a gas service regulator vent outlet (**G**).
- Not above a gas meter/regulator within 35 in. (.9m) horizontally of the vertical center line of the regulator (**G**).
- Not less than 3 ft. (.91m) below overhang (i.e. roof)

VENT LOCATION (CONT.)

Where passage through a wall, or partition of combustible construction is desired, the installation shall conform to CAN/CSA-B365. See specific venting instructions (page 9) for more information.

The vent **MUST NOT** connect to any air distribution duct or system, or exhaust into an enclosed or semi-enclosed area, but directly to the outside. Termination in a garage, attic, crawl space or other location in which fumes can accumulate must be avoided. The installation with the vent extending vertically through the roof of the building must have the proper termination cap installed. See specific venting instructions for more information.

Bixby recommends not installing a venting system in a pipe chase or permanent wall structure. Vent pipe should be accessible for annual inspection and maintenance. Because solid fuel produces ash, Bixby Energy Systems is not responsible for flue by-products that might damage or discolor building structures (walls, floors, roofs, etc.). If the vent is to be installed inside an existing masonry chimney or other unheated structure (or if excessive condensate forms), it is recommended that the pipe be insulated with non-combustible insulation (such as calcium silicate). See your local dealer for insulation options.

LOCATION OF EXHAUST

The exhaust is located on the back of the stove. See Figure 8 for dimensions.

NOTE: If a hearth pad is used, add the height of the hearth pad to the height of the exhaust hole location from the base, or measure from the top of the hearth pad with the pad in place.

NOTE: The shipping box can be used as a template for locating the center of the hole for the vent pipe in a horizontal venting configuration. Place the diagram of the top of the stove on the hearth pad and bend up the back side of the box.

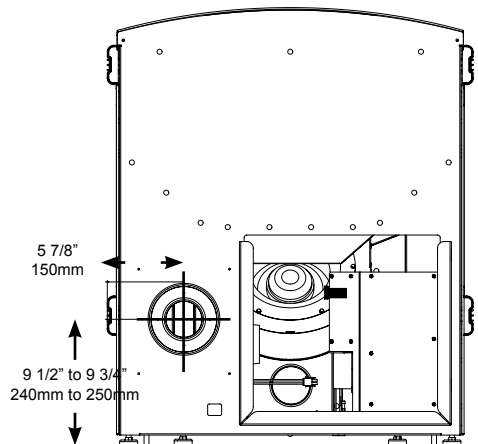


Figure 8

VENTING CONFIGURATIONS

The allowable venting configuration is to be 25 equivalent horizontal feet (7.5 m) or less as determined through the following table:

Type of Pipe	Qty		Factor (ft)	Total
90 deg elbow		x	5 (1.5m)	
45 deg elbow		x	3 (0.9m)	
Horizontal Pipe		x	2 (0.6m)	
Vertical Pipe		x	0.5 (0.15m)	
Altitude (in thousands of feet)		x	2.5 (0.8m)	
			Total	

Example:

Type of Pipe	Qty		Factor (ft)	Total
90 deg elbow	1	x	5	5
45 deg elbow		x	3	
Horizontal Pipe	1	x	2	2
Vertical Pipe	20	x	0.5	10
Altitude (in thousands of feet)	2	x	2.5	5
			Total	22

NOTE: Horizontal runs, should be minimized (48 in. (1.2m) maximum per section) to prevent the accumulation of ash. If possible replace horizontal offsets (two 90 degree elbows) with 45 degree offsets (two 45 deg elbows) to go around building structures.

GENERAL VENT INSTALLATION INSTRUCTIONS

Follow the Room Heater manufacturer's instructions and safety manual for maximum efficiency and safety.
Connect only BIXBY ENERGY SYSTEMS DIRECT VENT to BIXBY ROOM HEATERS.

CAUTION: Connect only one flue per appliance.

Do not install a flue damper in the exhaust venting system of this unit.

Do not connect this unit to a chimney flue serving another appliance.

Over firing (too much fuel) can damage the Room Heater, vent pipe, or surrounding structures.

- The use of three (3) sheet metal screws (self-tapping screws are recommended) is required at every connection. Do not use longer than 3/4 in. (20mm) screws to prevent damage to the inner pipe. The indents in the pipe will assist in the placement of the screws.
- **High temperature (500F/260C) silicone sealant is required on all 3 in. diameter inside joints.** Either high temperature or clear silicone sealant is recommended for the outer 5 in. outside joint. A 1/8 in. (3mm) bead, 1/2 in. (12mm) from the male (cripped) end of both sections of pipe is recommended. See Figure 9.
- The joint is not sealed unless the seam between the two outer sections of pipe is tight and the indent locks into the groove. See Figure 10.

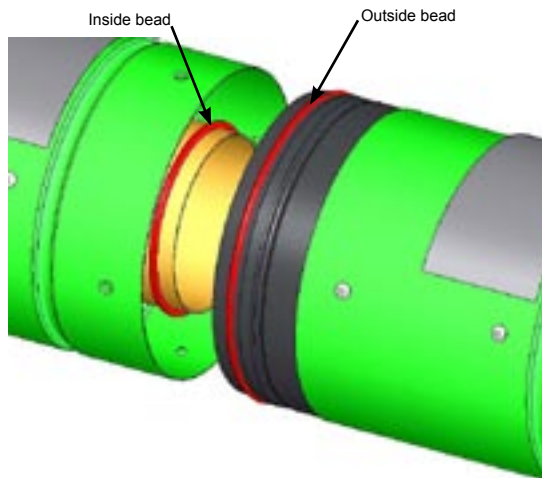


Figure 9

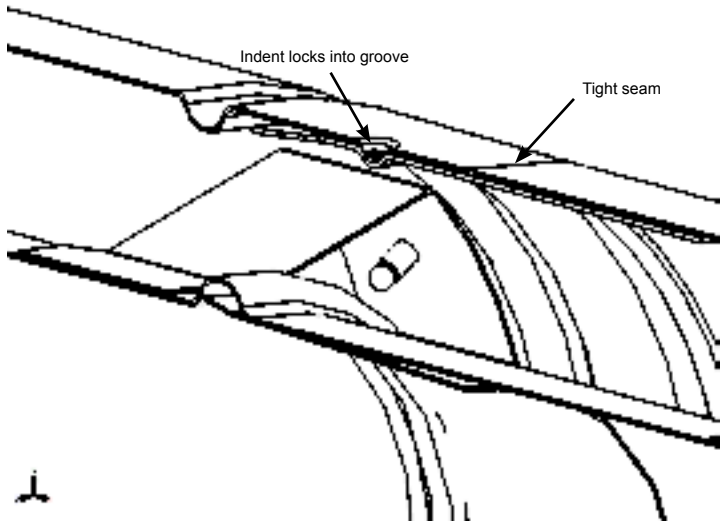


Figure 10

WARNING: ALWAYS MAINTAIN AT LEAST 2 in. (61mm) CLEARANCE FROM COMBUSTIBLE MATERIALS.
Where the chimney passes through additional floors and ceilings, always install a Firestop/Support.

VERTICAL INTERIOR INSTALLATION

NOTE: Vertical exterior installation is not recommended due to excessive condensation.

NOTE: Bixby recommends not installing a venting system in a pipe chase or permanent wall structure (other than to exit the room using the Wall Thimble). Vent pipe should be accessible for annual inspection and maintenance. Bixby is not responsible for damages to pipe chases, or other wall structure resulting from vent failures.

The flue exit is on the back of the Room Heater and if vertical interior installation is desired:

1. Place the Room Heater according to the installation instructions. See Figure 7 for vent locations.
2. Connect a Starter Collar to the back of the Room Heater using three screws. See Figure 11.
3. Insert an Elbow through the Starter Collar, into the back of the Room Heater. Align and push together until snapped into locked position. See Figure 11.
4. The outer pipe sections should not have a visible gap. See Figure 10.
5. Drop a plumb bob to the center of the Elbow connected to the Room Heater flue outlet and mark center point on the ceiling, cut and frame a 10 in. x 10 in. (25cm x 25cm) square hole in the ceiling for the Ceiling Support Firestop. See Figure 12.
6. Connect the vertical pipe to the elbow attached to the Room Heater: Attach each section of pipe by pushing male and female ends of pipe together until pipe is in locked position. See General Vent Installation Instructions, page 14.
7. Install Wall Straps every four (4) feet in both vertical and horizontal vent runs (required). See Figure 13.
8. When the pipe passes through the Firestop/Support at ceiling, attach the vertical pipe support around pipe and let it rest on the topside of the Firestop/Support. If desired, a Decorative Collar can be installed over the Firestop/Support for a finished look.
9. After lining up for hole in roof, use same method as above and cut either a round or square hole in the roof. Always cut the hole at least 4 in. (10cm) larger than the diameter of the pipe. Install the upper edge and sides of Flashing under the roofing materials and nail to the roof along the upper edge. Do not nail the lower edge. The Vent Cap must extend a minimum of 36 in. (91cm) above the roof. Also, the Vent Cap must be 24 in. (64cm) higher than the highest part of the roof or structure that is within 10 ft. (3m) of the chimney, measured horizontally. See Figure 13.
10. Seal all nail heads with high-temperature waterproof sealant. To finish, apply high-temperature waterproof sealant where the Storm Collar will meet the vent and Roof Flashing; slide Storm Collar down until it sets on the Roof Flashing, tighten the bolt on the Storm Collar, put the Vent Cap on and push to lock. See Figure 14.

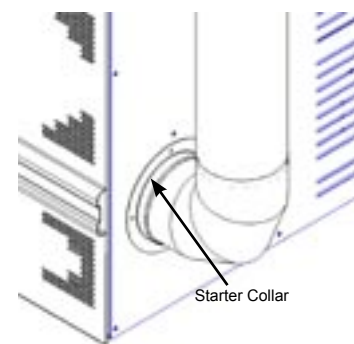


Figure 11

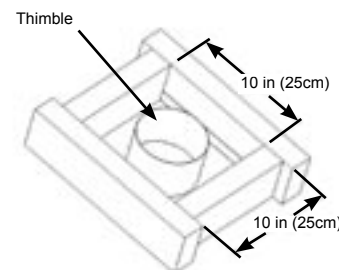


Figure 12

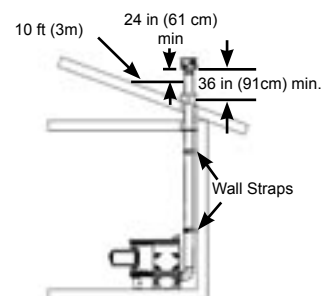


Figure 13

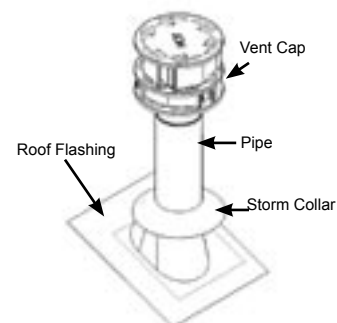


Figure 14

HORIZONTAL THROUGH-THE-WALL-INSTALLATION

The flue exit is on back of the Room Heater, and if a horizontal through-the-wall installation is desired:

1. Place the Room Heater according to installation instructions. See Figure 7 for vent locations.
2. Locate vent termination per the NFPA rules. See Figure 7 (page 11).
3. Cut and frame a 10 in. X 10 in. (25cm x 25cm) square opening in the wall (the hole in the wall board may need to be larger). See Figure 12 (page 15). The telescoping Wall Thimble is to be installed in the interior and exterior portion of the wall. (Wall Thimble adjusts from 4 1/4 in. to 7 in. (11cm to 18cm) to accommodate most wall thicknesses). See Figure 16.
 - The Wall Thimble flanges are to be attached to the supports with four screws, one in each corner. Use of 1/2 inch or longer screws are recommended.
 - The Wall Thimble may be covered on the inside by the Decorative Collar, see venting parts list for color options. It is important to slide the collar on before adding additional venting.
4. Attach the outside vent flashing to the exterior wall with four screws. Seal the outside section of the Horizontal Flashing and Wall Thimble to the exterior wall with non-hardening sealant. Top tab is to be placed under wall siding.
5. Connect the Starter Collar to back of Room Heater and attach sufficient pipe to the Room Heater to extend 4 in. - 8 in. (10-20cm) beyond the Flashing. See Figure 16.
6. Anchor the pipe to the Horizontal Flashing with the three brackets using the self tapping sheet metal screws shipped with the flashing. The back/base of the cap is to extend 6 - 12 inches (15-30 cm) from the flashing. See Figures 15 and 17.
7. Attach the Vent Cap, note orientation of embossment, to the extended pipe. Use high temperature (500 F/260 C) silicone sealant between all pipe joints. See Figure 17.

NOTE: Some amount of soot and discoloration may occur on the outside of the house.

NOTE: If ice forms on the Vent Cap in cold weather, run the Room Heater on high for a couple of hours to clear.

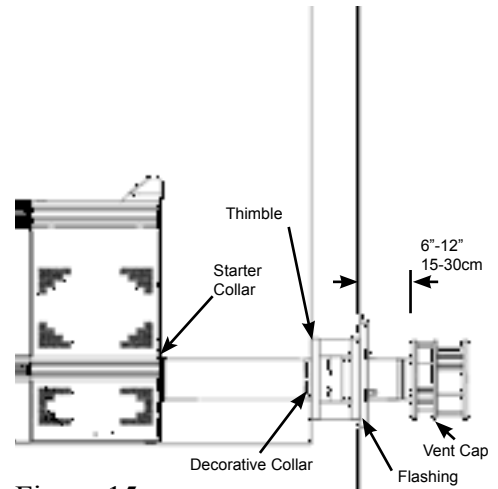


Figure 15

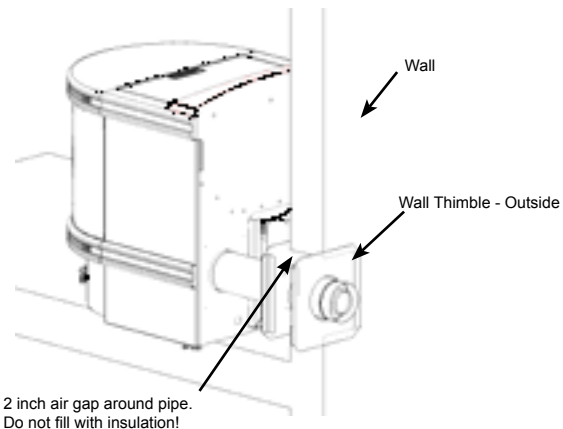


Figure 16

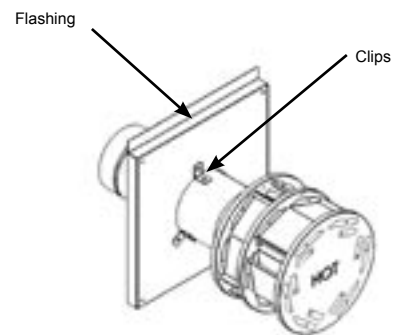


Figure 17

INSTALLATION IN A CATHEDRAL CEILING

1. Mark a line on the side of the Cathedral Ceiling Support Box to correspond to the line of the roof pitch. Allow for the Support Box to protrude below the finished ceiling a minimum of 2 in (5 cm). See Figure 18.
2. Position the appliance at its proper location on the floor. Pay close attention to the installation instructions as to distances from combustibles, etc. The appliance must also be positioned so the Support Box will not interfere with roof rafters or other structural framing.
3. Run a plumb line from the center of the flue exit on the Room Heater to the ceiling. Mark the point on the ceiling where the plumb line intersects. This represents the center of the support box. Drill a small hole through the ceiling at this point, so it can be located from the top of the roof.
4. From the roof, locate and mark the outline of the Support Box.
5. Remove shingles or other roof covering as necessary to cut the rectangular hole for the Support Box. Cut the hole 1/8 in (3 mm) larger than the dimensions of the Support Box. The rectangular hole should be centered on the small hole which you drilled through the ceiling to mark the location. Again, insure that you are not cutting through rafters or framing members.
6. Run the Support Box through the roof, and place it so that the bottom of the Support Box protrudes at least 2 in (5 cm) below the finished ceiling. Align the Support Box vertically and horizontally with a level. Temporarily tack the Support Box in place through the inside walls and into the roof sheathing. See Figure 19.
7. Using tin snips, cut the Support Box from the top corners down to the roof line, and fold the resulting flap over the roof sheathing. Before nailing it to the roof, run a bead of non-hardening sealant around the outside top edges of the Support Box so as to make a seal between the box and the roof. Clean out any combustible material or debris from inside the Support Box. See Figure 20.

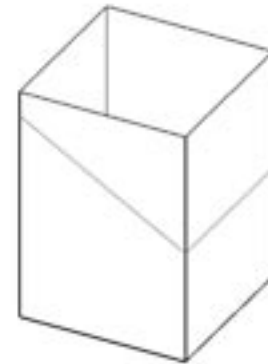


Figure 18

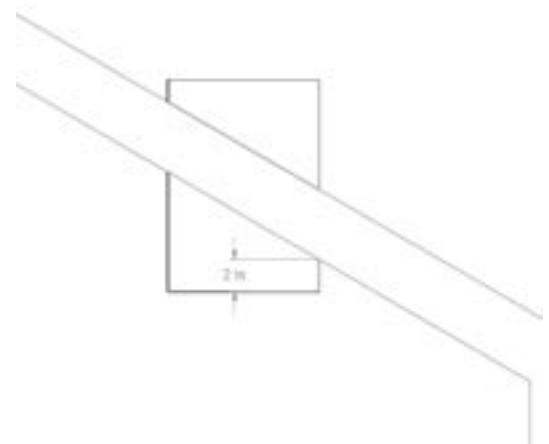


Figure 19

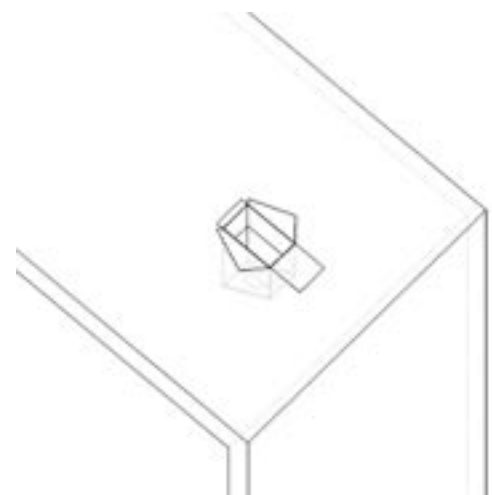


Figure 20

INSTALLATION IN A CATHEDRAL CEILING (CONT.)

8. Place the Vertical Support Clamp (5000576), around the top of the hole in the bottom of the Support Box. See Figure 21.
9. Connect the necessary amount of pipe sections to reach the Room Heater at a point where the Vent Cap will extend a minimum of 36 in (91 cm) above the roof. Also, the cap must be 24 in (61 cm) higher than the highest part of the roof or structure that is within 10 ft (3 m) of the chimney, measured horizontally. See Figure 22.
10. Run the assembled pieces of vent pipe down through the Clamp of the Support Box and connect to the appliance with an Elbow and Starter Collar. See General Vent Installation Instructions for details on connecting the pipe together.
11. Using a level, make slight adjustments in the position of the appliance until the pipe is truly vertical. Tighten the bolts in the Clamp.
12. Slip the Flashing over the pipe section(s) protruding through the roof. Secure the base to the roof with roofing nails. Insure that the roofing material overlaps the top edge of the Flashing. See Figure 22.
13. Seal all nail heads with high-temperature waterproof sealant. To finish, apply high-temperature waterproof sealant where the Storm Collar will meet the Vent and Flashing; slide Storm Collar down until it sets on the Flashing, put the Vent Cap on and push to lock. See Figure 14 (page 16).

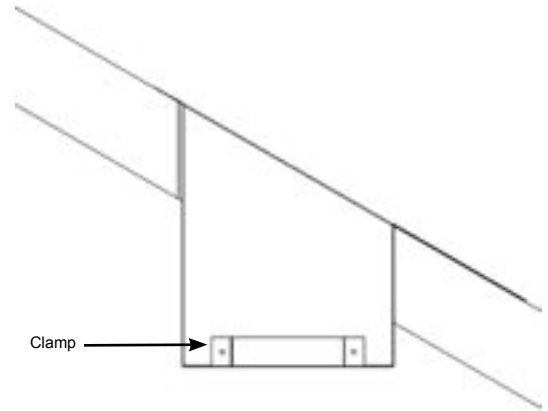


Figure 21

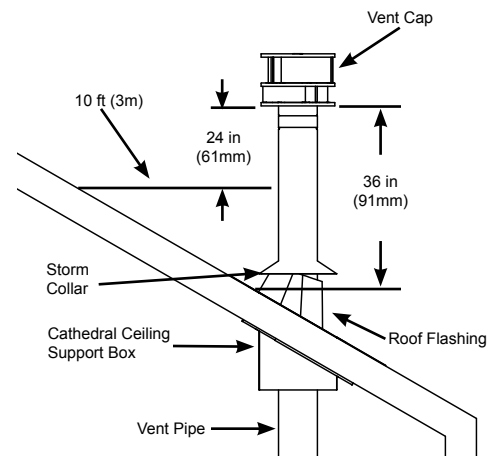


Figure 22

CONNECTING TO EXISTING CLASS A CHIMNEY EITHER 6" OR 8"

Due to cleaning, inspection and condensation issues, Bixby no longer recommends connecting the Room Heater to an existing Class A chimney.

INSTALLATION THROUGH THE SIDE OF A MASONRY CHIMNEY

Due to cleaning, inspection and condensation issues, Bixby no longer recommends connecting the Room Heater to an existing masonry chimney. The venting may pass through the chimney to the outside, see previous page for horizontal venting information.

LOG PLACEMENT INSTRUCTIONS

The use of the logs provided with the Room Heater is optional.

1. Locate the log box and remove the three logs. See Figure 23.
2. Place one of the “Y” logs on the right side of the burn pot with the small end pointing towards the back of the firebox. See Figure 24.
3. Place the second “Y” log parallel to the front of the Room Heater with the small end inside the first “Y” log. See Figure 25.
4. Place the straight log on the left side of the burn pot with the end inside the “Y” of the second log. See Figure 26.



Figure 23

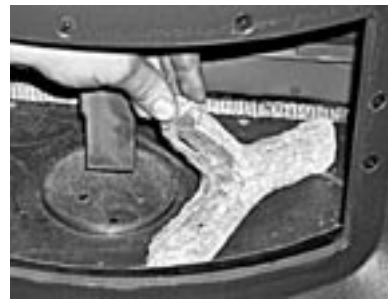


Figure 24



Figure 25

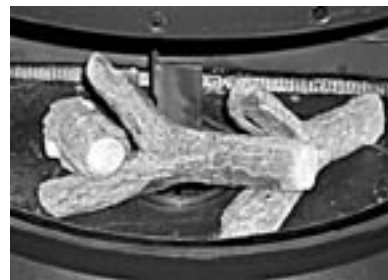


Figure 26

OPERATING INSTRUCTIONS

FUEL

It is strongly recommended that only Dry Shelled Corn (whole kernels), Wood Pellets (recommended maximum size approximately ¼ in (6mm) diameter x 1 ¼ in (32mm) long), Bixby Certified Corn or approved pelletized biomass fuel be used in this Room Heater (Up to 8,700 BTU's per pound). When using Dry Shelled Corn, clean corn (free of foreign material like cobs, twigs, stones, etc.) with moisture level around 14% or less is recommended. The use of “cracked corn” or seed corn is not recommended due to combustion and feed issues. Use Premium Wood Pellets made for use in Pellet Room Heaters. High-grade wood pellets may be used, but cleaning of Bixby Room Heater and venting components may be required more frequently due to excess ash resulting from fuel contamination from pitch and other impurities in the wood.

Caution should be used in burning unclean or high moisture corn. Corn that is unclean can produce undesirable deposits and excessive ash in your Room Heater and heating efficiency can be dramatically affected. Corn of more than 14% could generate mold in the hopper plus increase issues of condensation in the vent system. Use of unsuitable fuel may harm the Bixby Room Heater and could void its warranty. High performance equipment, like your Room Heater, performs better when given the best fuel.

The Room Heater is setup to burn corn from the factory. To burn wood pellets, simply open the left side pannel and locate the fuel switch. Toggle the switch from “CORN” to “WOOD”. See Figure 27.

CAUTION: DO NOT BURN GARBAGE, OR FLAMMABLE FLUIDS SUCH AS GASOLINE, NAPTHA OR ENGINE OIL IN YOUR BIXBY ROOM HEATER. Store any fuel away from your heater.

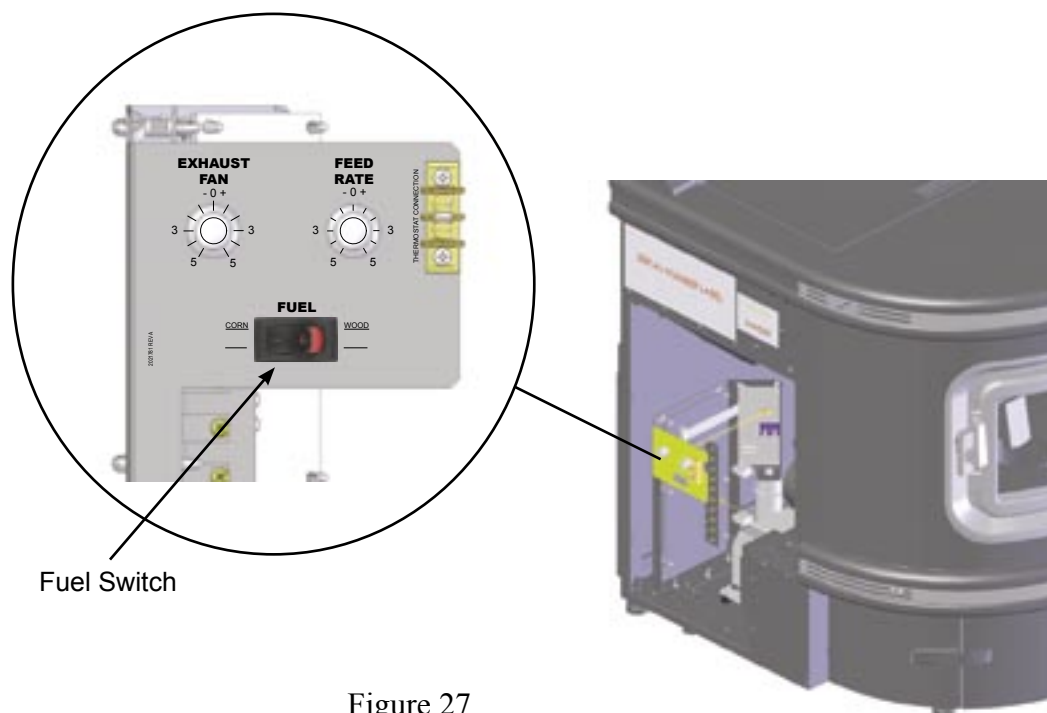


Figure 27

START UP

NOTE: It is recommended that all gold or silver plated surfaces of the Bixby Room Heater be cleaned with a mild glass cleaner before initial startup.

NOTE: Use of a surge protector is recommended but not required for heater operation.

CAUTION: Operate the Bixby Room Heater only with the Fuel Hopper Door closed. Failure to do so may result in emission of products of combustion from the Fuel Hopper under certain conditions. Do not overfill or extend this hopper.

1. Open the Fuel Hopper Door on the top of the Bixby Room Heater and fill the hopper with Bixby Certified Corn or Biomass Pellet Fuel, or an approved alternative. The capacity of the fuel hopper is approximately 106 lbs (48 kg).
2. Verify that the fuel deflector is in place.
3. Plug in the Bixby Room Heater's grounded, 3-prong power cord. A grounded 115 volt, 60 Hz, 15 amp power supply is required. **Do not use extension cords.** The furnished power cord must be plugged directly into a suitable outlet. Route power cord away from the Bixby Room Heater. The exhaust fan will operate for approximately 30 minutes when the heater is plugged in. This is normal and is to ensure that any remaining combustion gases are removed in the event of a power outage.
4. Manual Operation: Press the "On" button on the Control Panel. Then press the up/down arrows to bring heat output to the desired level (the "Up" arrow increases the heat output, the "Down" arrow decreases it). The first time the Bixby Room heater is used, or in cases when fuel in the hopper has been completely expended, sufficient fuel may not have been supplied to allow the unit to start properly. If this happens, press the "Off" button, and then the "On" button to restart the unit after fuel begins feeding into pot.
- 4a. Optional Thermostat Operation: The thermostat will not start the Room Heater. When the thermostat is calling for heat, the Room Heater will operate at the selected level. When the thermostat is not calling for heat, the Room Heater will drop to level 1 to maintain the burn. The lights will blink when the thermostat is not calling for heat.
- 4b. Manual Igniting Information: To manually ignite the Room Heater, press the "ON" button on the control panel. After the fuel has started to be delivered to the burn pot, remove the ash drawer to access the bottom of the pot. Ignite the fuel by heating the bottom of the burn pot with a MAP torch and maintain a flame. Once the fuel has ignited, replace and close the ash drawer and verify the flame is stable. Wood pellets and starter gel may be used to assist the lighting. See below for cautions.
5. Normally, the fuel will ignite within three to seven minutes, and full flame stability will be achieved within twenty minutes. Heat production will then begin to reach the desired comfort level. If stable ignition does not take place within this time and there is fuel building up in the burning area, press the "Off" button and then the "On" button to restart the unit.
6. To shut down the Bixby Room Heater, press the "Off" button on the control panel, allow three to ten minutes for flame-out to occur.

NOTE: The convection blower and the exhaust fan will continue to run for 30 minutes as the Room Heater cools.

7. Power Outages: If there is a momentary (less than 10 second) loss of power, your heater will return to the set heating level. If the power is interrupted for longer than 10 seconds, you will have to restart your unit after the power is restored. The #1 light will be blinking, indicating a power failure.

NOTE: Ash Drawer: Opening the ash drawer during heater operation will disable the dump process. Be sure to replace and latch the ash drawer once emptied. If the ash drawer is open for more than 20 minutes, the unit will go directly into shut-down mode.

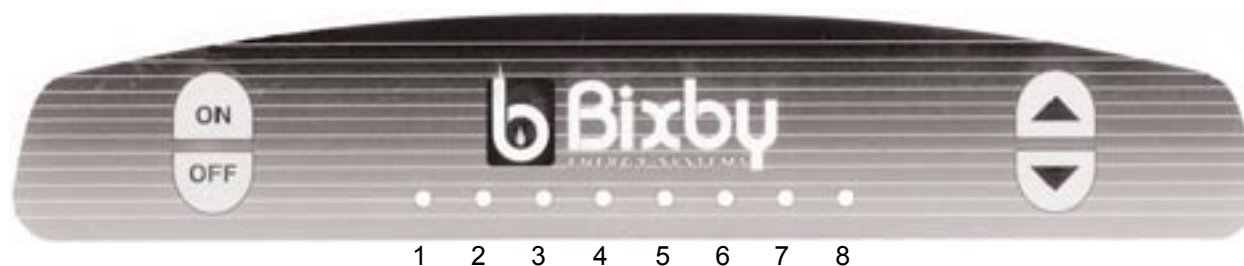
NOTE: Front Door: The door must be closed and latched while the Room Heater is running. Always allow the Room Heater to cool before opening for cleaning. If the door is open for more than one (1) minute, the unit will go directly into shut-down mode.

CAUTION: Never use highly-volatile fuels such as gasoline, gasoline-type lantern fuel, kerosene, charcoal lighter fluid, or similar substances in the Bixby Room Heater. Keep all such liquids well away from the Bixby Room Heater when it is in use. Use of such fuels could cause over-firing and will void the unit's warranty.



Wood pellets and starter gel may help if there is difficulty in manual starting, but do not use starter gel after startup.

FLASHING INDICATOR LIGHT SYSTEM



The 8 heat level indicator lights on the Bixby Room Heater are also designed to act as potential problem indicators. Whenever one of these lights begins to flash, consult the chart below for reason and recommended solution.

Light #1	Power failure during Room Heater operation: Push the “Off” button and then the “On” button to restart.
Light #2	Operating temperature not reached: Push the “Off” button and then the “On” button to restart. Verify fuel level in the hopper.
Light #3	Exhaust system or hopper area is overheating. Reduce the feed rate (see Troubleshooting section). If problem continues, contact your Bixby Dealer.
Light #2 & #3	Empty hopper or potential blocked flue warning. Check hopper for fuel. If fuel is present, flue (exhaust) may be blocked. Clean the exhaust portion of the Room Heater and the attached venting. See Maintenance and Cleaning Procedures for details. If problem continues, contact your Bixby Dealer.
Light #4	Room Heater door is ajar (open). Make sure the door is tightly closed. If the door is open for more than one minute, the Room Heater will shut down. Contact your Bixby Dealer if problem continues.
Light #5	Ash Drawer is ajar (open): Make sure the ash drawer is tightly closed. The Room Heater will not start if the ash drawer is not closed. If the ash drawer is open for more than 20 minutes, the Room Heater will shut down. Close ash drawer and restart the Room Heater.
Light #6	Exhaust fan failure. Contact your Bixby Dealer for service.
Light #7	Fire pot mechanical malfunction: Push the “Off” button and then the “On” button to restart. If malfunction continues, clear jam in burn pot once cool.
Light #7 & #1	Left igniter failed. Contact your Bixby Dealer for service. The Room Heater will continue to operate, but will take longer to ignite from a cold start.
Light #7 & #2	Right igniter failed. See above. If both igniters have failed (Lights 7, 1 & 2), the Room Heater will not be able to autostart. See Manual Lighting instructions (previous page).
Light #7 & #1-3	Internal error - possible ignitor electrical fault - unplug Room Heater and contact your Bixby Dealer for service.
Light #8	Feeder wheel failure. Open hopper lid, remove feeder wheel cover, clear jam and restart Room Heater. If this continues, contact your Bixby Dealer for service.

MAINTENANCE AND CLEANING PROCEDURES

The Bixby Room Heater has been designed to reduce the frequency of cleaning. However, if the simple maintenance tasks detailed below are not performed - the Room Heater will not run properly and will lead to reduced heating efficiency, increased operational problems and potential harm to its components. Reading and following the maintenance instructions will keep the Room Heater running at optimal efficiency and will help to reduce problems over the long run. Many problems can be solved by simply cleaning the Room Heater. This should be the first step in troubleshooting if the Room Heater is not working properly.

NOTE: Keep the area around the Room Heater clean and free of dust and debris.

CAUTION: Always be sure to allow the Bixby Room Heater to cool and then unplug before any maintenance procedures are performed.

COMPONENTS REQUIRING PERIODIC CLEANING

FIRE POT/BURNER

This device is powered by an electric motor. The automatic operation serves to clear the ash/deposits from the Fire Pot.

The sidewalls of the fire pot should be brushed monthly with the included brush to remove deposits. If needed the pot may be removed and soaked in a bucket of water. To remove the burner, remove the fuel deflector, remove the top burner plate (handle in front of the burner), remove the two clips holding the burner in place and lift out the burner.

Good combustion is dependent on delivering the right amount of air to the burn pot. Keeping the pot clean and the holes clear will keep the air flowing to the fuel and reduce fuel related problems.

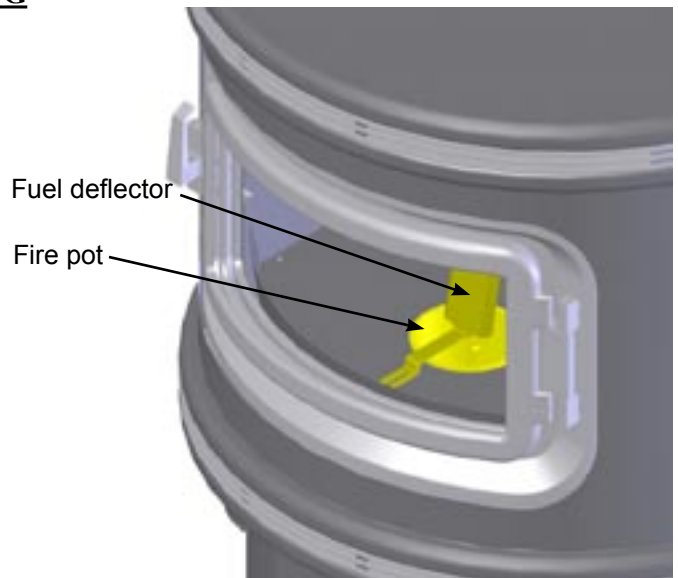


Figure 27

ASH DRAWER

This container should be emptied as needed. With the Bixby Room Heater simply undo the latches, pull the Ash Drawer out and empty its contents.

CAUTION: Drawer and contents may be hot.

NOTE: Return drawer promptly (less than 20 minutes) after emptying. While the drawer is open, the Room Heater will not dump the ashes and could postpone a cycle leading to build up in the burner pot.

The ash drawer should be checked / emptied after burning approximately 100-200 lbs. (1 - 2 full hoppers) of fuel. Actual frequency will vary with usage, check every few days.

Ashes should be placed in a metal container with a tight-fitting lid. The closed container of ashes should be placed on a non-combustible floor or on the ground, well away from all combustible materials, pending final disposal. If the ashes are disposed of by burial in soil or otherwise locally dispersed, they should be retained in the closed container until all cinders have been thoroughly cooled.

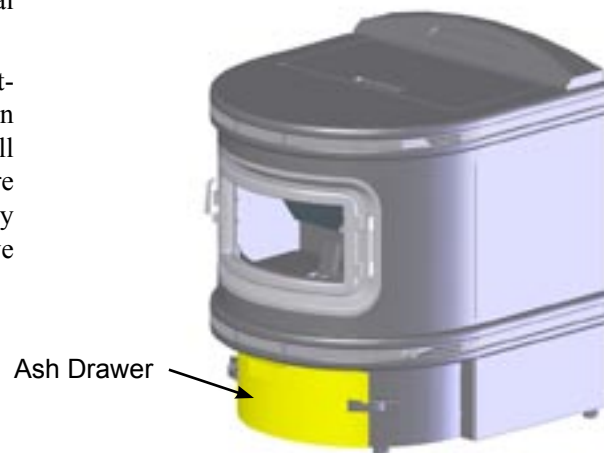


Figure 28

HEAT EXCHANGER TUBES

Fly ash and other deposits will accumulate on the Heat Exchanger Tubes over time and will act to insulate the tubes, reducing their ability to heat the convection air and thus the home or building.

There are two banks of Heat Exchanger Tubes, both of which are equipped with a cleaning lever. The levers are located above the back ceramic pannels. To clean the tubes, pull the right side lever straight up approximately 1" (2 cm). Carefully remove the back panel. Repeat for the left side. Once the back panels have been removed, slide the levers up and down untill they move freely. Vacuum out the base of the Heat Exchanger Tubes and the firebox. See Figure 29.

To replace the back panels, lift the left lever straight up slide the left panel underneath. Repeat for the right hand side. See Figure 30.

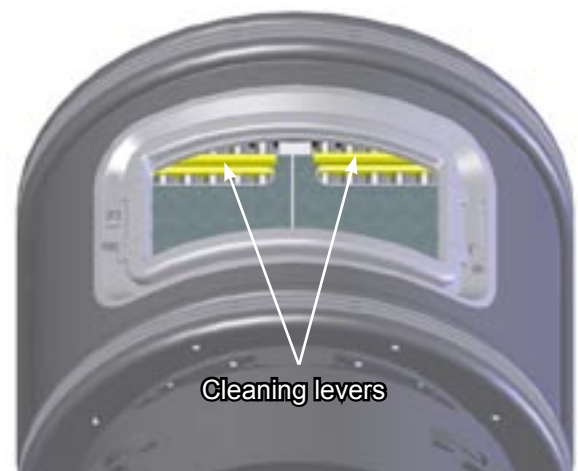


Figure 29

SOOT AND FLY-ASH FORMATION AND NEED FOR REMOVAL

The products of combustion will contain small particles of fly ash. The fly ash will collect in the exhaust venting system and restrict the flow of the flue gases. Incomplete combustion, such as occurs during start-up, shutdown or incorrect operation of the heater will lead to soot formation which will collect in the exhaust venting system. Inspect the exhaust venting system at least yearly, cleaning as required. Also, inspect any exhaust outlet to maintain a clear and unobstructed exhaust/intake port.

CAUTION: Use only an approved ash vacuum for this purpose.

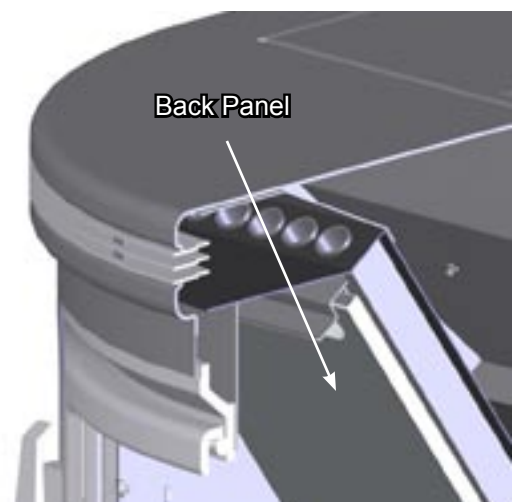


Figure 30

EXHAUST FAN

CAUTION: Unplug the Room Heater from its power source before cleaning the impeller.

To access this component, open Right Side Panel, and remove the fan cover. Using a brush and a vacuum, clean the impeller. Inspect gasket. If it is torn, replace gasket before placing unit in service.

This cleaning should be performed once a year.



Figure 31

CLEANING THE EXHAUST MANIFOLD

The Exhaust Manifold should be cleared on a yearly basis to ensure proper air flow in the Room Heater.

1. Turn off the Room Heater and allow to completely cool (minimum of 2 hours).
2. Remove the back panels (see previous page for instructions).
3. Move the cleaning levers up and down several times until they move freely.
4. Using a shop vac, clean the ash around the base of the tubes.
5. Use the Exhaust Manifold Cleaning Kit (available at your dealer) to clean in behind the tubes.
6. Replace the heat exchanger cover plates.



Figure 32

FUEL HOPPER

The Fuel Hopper should be cleaned out on a regular basis (monthly) to remove fines and ensure proper fuel delivery.

1. Remove the hopper cover and set aside.
2. Empty the hopper of as much fuel as possible by either running the Room Heater, shop vac, or by hand.
3. Vacuum the base of the feeder wheel to remove small particles and other debris.
4. Remove the feed wheel access plate and vacuum around the feed wheel.

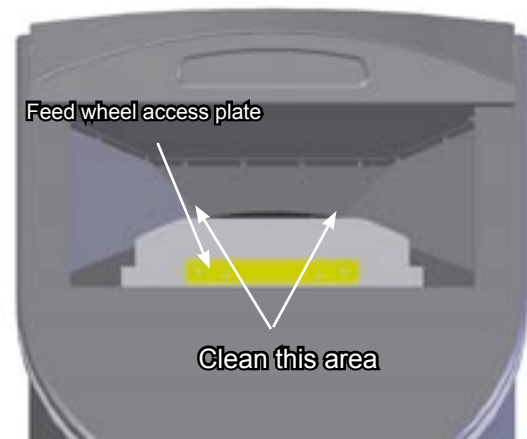


Figure 33

CLEANING THE GLASS

Depending on the fuel used and the amount of use, the glass window will need to be cleaned. The glass can be cleaned using any available glass cleaner. Do not use abrasive cleaners to clean the glass. For tough cleaning use cleaners designed for ceramic stove tops.

CAUTION: Before cleaning the glass, allow the Room Heater to cool (minimum 2 hrs). Do not spray window cleaner on the glass if hot.

If the glass needs to be cleaned more than every 4 to 5 bags of fuel (two full hoppers), contact your dealer to verify Room Heater operation.

NOTE: Tighten door latch after the first month of operation to maintain door seal.

CLEANING THE FEEDER TUBE

The Fuel Feeder Tube should be cleaned out on a regular basis (monthly) to ensure proper fuel delivery.

1. Allow Room Heater to cool, minimum 2 hrs.
2. Open the Glass Door.
3. Using a screw driver or small brush, scrape the inside of the feeder tube.

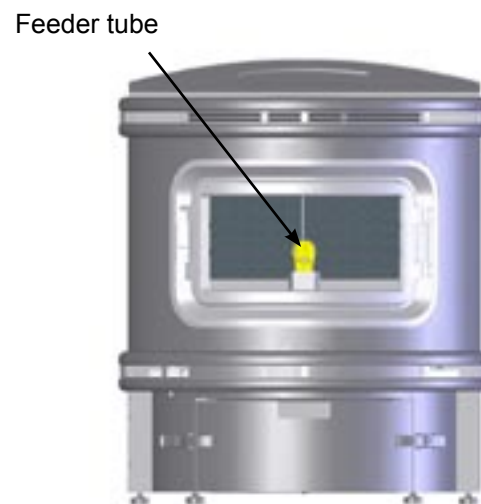


Figure 34

If the tube needs to be cleaned more than every 4 to 5 bags of fuel (two full hoppers), clean the fuel hopper (above).

CLEANING THE LOWER PADDLE

The Lower Paddle needs to be cleaned once a month in order to operate efficiently. The ash dumping process causes the holes in the lower paddle to eventually clog with ash.

1. Make sure the Room Heater is off and cool.
2. On the control panel, push the “ON” and then the “OFF” button immediately to clear the burn pot and to place the lower paddle in the correct position for cleaning.
3. Unplug the Room Heater.
4. Open the glass door or ash drawer to gain access to the burn pot.
5. Looking into the burn pot, insert a 1/8” drill bit, toothpick, or similar diameter tool into the holes, clearing each blocked hole.

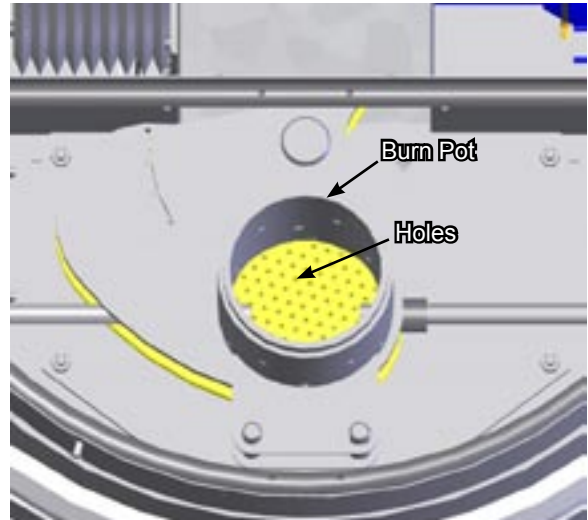


Figure 35

CLEANING THE AIR FILTER FOR THE IGNITION AIR COMPRESSOR

The air filter for the air compressor will only need to be cleaned if the Room Heater is installed in a dusty environment. Inspect filter annually. If it appears dirty, remove and clean with a mild detergent and water. Once dry, replace.

REPLACING THE ROOM HEATER FILTER

The air filter for the Room Heater is located on the back of the Room Heater. The filter should be checked monthly and replaced every 2 to 3 months depending on usage. A clogged filter will reduce the efficiency of the Room Heater. The filter can be replaced with most 16” x 16” x 1” filters, or if not available, 20” x 16” x 1” filters. Do not run the Room Heater with out the filter in place as it’s primary function is to keep dust and dirt out of the mechanical components in the Room Heater.

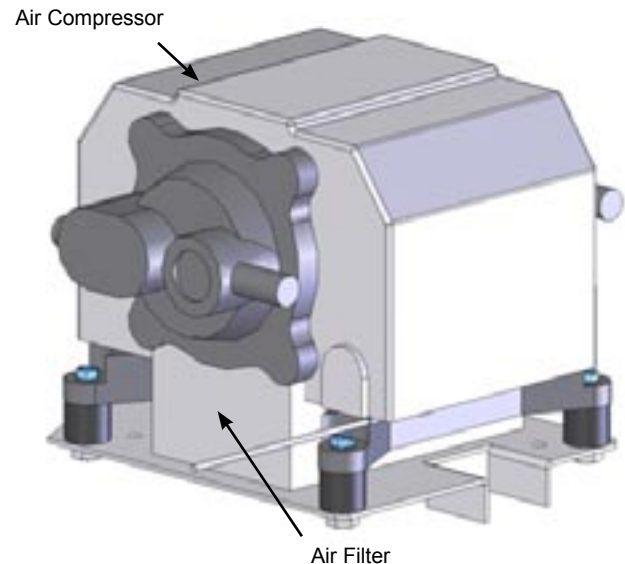


Figure 36

TROUBLESHOOTING PROCEDURES

When experiencing a malfunction, please read through the following list of possible problems to determine which of them most closely approximates the trouble you are encountering. This will reduce troubleshooting time and effort.

CAUTION: Always be sure to unplug electrical connection to Bixby Room Heater before any service procedures are performed.

NOTE: If you need to call the service department, verify the serial number (located on the left side of the hopper) has been recorded on page 1 and have this manual with you.

NOTE: Many performance issues can be solved through the use of the trim pot knobs located near the control board behind the right side panel. Manual adjustment of either the fuel feed rate or the exhaust fan speed may be necessary with different fuels (change in moisture for example), different seasons (change in outside air temperature) or other changes in the operating conditions. Read through the troubleshooting chart below to determine which knob to adjust and in which direction.

<u>PROBLEM</u>	<u>POSSIBLE CAUSE/REMEDY</u>
No response when Room Heater is plugged in.	Check the 120 volt power supply circuit for adequate voltage. Check fuse on circuit board and replace if necessary.
No fire or fuel in burn pot.	Check to assure that there is fuel in the hopper. Verify feeder wheel is turning. (See error codes) Check to assure that the fuel feeder tube is not blocked by fines or “bridging” of fuel pellets or kernels. Push the “Off” button and then the “On” button to attempt a new startup. Door or ash drawer is ajar (See error codes).
Flame goes from high to low to high	Feed rate is low, turn the feed rate up 1 notch using the Fuel trim pot knob and observe. Repeat if necessary.
Smoke from convection air outlet (louvers) or hopper.	Some wisps of smoke may occur during startup. This is normal. Check for debris under the feeder wheel. If the smoke doesn’t clear within 15 minutes or if excessive immediately shut down unit, ventilate the area and contact your dealer.
Unburnt fuel in ash drawer	A small amount is normal. During start up, the burner pot is emptied and as a result, unburnt fuel left in the pot will be dumped. Excessive ammounts is likely due to fire going out due to either too much fuel (smothering the fire) or too much air (burning too fast). Observe Room Heater, if pot begins to fill with fuel, turn fuel down. If fuel is burning too fast, turn air down. Hopper may have run out of fuel causing the fire to go out.
Ash dump jamming on clinker	Burn may be too hot. Feed rate / fan speed may need adjustment - turn fan speed down or feed rate up 1 notch and observe. Repeat if necessary.

TROUBLESHOOTING CONT.

<u>PROBLEM</u>	<u>POSSIBLE CAUSE/REMEDY</u>
Fire fails to start	Be sure enough fuel is in the burn pot. Fuel should cover the bottom to a level of approximately 1/2". Check bottom of burn pot for built-up ash or deposits and remove. Check for flashing light #6, exhaust fan failure (contact your dealer). Push the "Off" button on the control panel. Wait three minutes and push the "On" button to attempt a new startup. NOTE: When the Room Heater is cold (below normal room temperature), a "second start" may be required. This is normal and a result of needing to pre-heat the burn pot. Check the igniter fuses located on the control board. (See error codes). Verify air pump is working during start up (feel pump for vibration). Inspect Room Heater and venting to see if cleaning is needed.
Fire starts but shuts down after approximately 12 minutes. (#2 indicator light)	Check feeder chute to be sure fuel is feeding. Check to make sure exhaust thermocouple wire has not been disconnected from the control board. (Two wire connection, upper right hand side). If feeder stops after 12 minutes, electronics may be faulty; contact your dealer for service.
Dark yellow and or lazy flame, slow start-up or excessive smoke present.	Check for spent fuel or too much fuel clogging bottom of burn pot. Burn pot, lower paddle holes may require cleaning. Heat exchanger tubes may require cleaning. Check vent pipe for blockage of exhaust gasses and incoming combustion air. Feed rate / fan speed may need adjustment - turn fan speed up or feed rate down 1 notch and observe. Repeat if necessary.
Excessive volume of fire, build-up of ash and dirty glass	Burn pot may require cleaning The venting system or heat exchanger may need to be cleaned. If the glass needs to be cleaned more than every 4 to 5 bags of fuel (two full hoppers), contact your dealer to verify Room Heater operation.
Burn pot filling with fuel	Room Heater may not be level Feed rate / fan speed may need adjustment - turn fan speed up or feed rate down 1 notch and observe. Repeat if necessary. Verify that the exhaust is not obstructed. See Maintenance and Cleaning instructions - Cleaning Heat Exchanger Tubes. In addition, the vent pipe may need to be cleaned. Holes in burn pot may need to be cleaned Check the alignment of the burn pot components Check the installation of the plates covering the heat exchanger tubes.

WIRING DIAGRAM

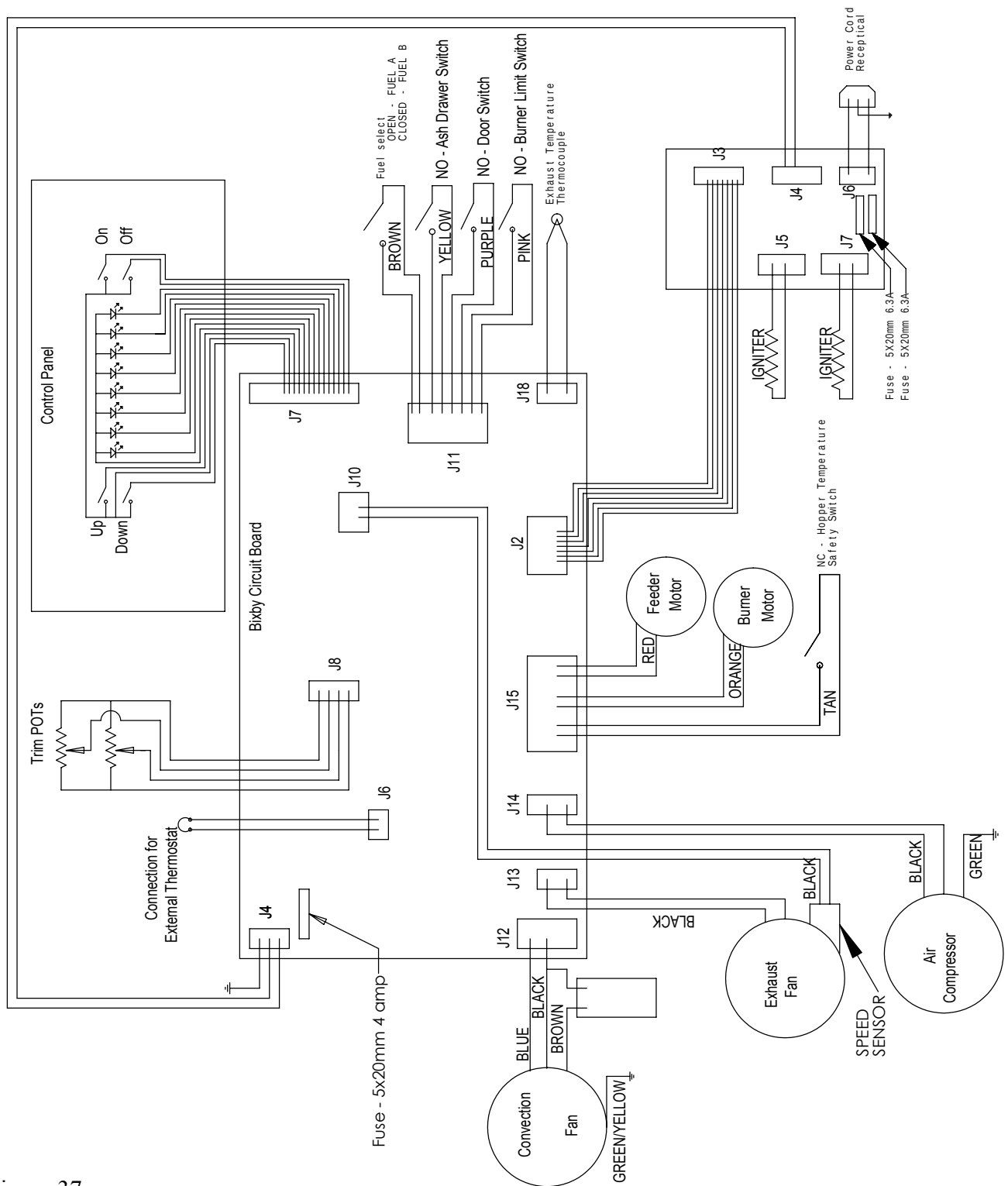
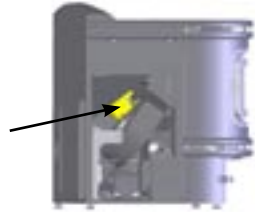
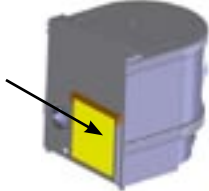


Figure 37

SERVICE PARTS LIST

Item Number	Description	Part	Location
4000080	Igniter		
4000121	Air pump		
4000113	Burner drive motor assembly		
4000006	Feeder wheel		
4000098	Feeder wheel motor assembly		
4000014	Convection fan capacitor		

Item Number	Description	Part	Location
4000022	Heat exchanger cover plate - Right Heat exchanger cover plate - Left		
4000048	Air Filter		
4000072	Control pad		
4000064	Main control board		
4000105	Exhaust fan		
4000056	Exhaust fan gasket		

Item Number	Description	Part	Location
4000353	Burner top plate		
4000361	Feed wheel sensor		
4000379	Fuel deflector		
4000329	Thermocouple		
4000337	Door Hinge		
4000345	Door Latch		
4000147 4000155 4000163	Door, Gold Door, Nickel Door, Black		

OPTIONAL ACCESSORIES

Item Number	Description
2013332	Kit, Vacuum Cleaning Accessory
2012219	Thermostat, Remote

VENTING PARTS LIST

Description	Color / Size	Part #	
Vertical Termination Cap	Black	5001368	
Wall Thimble	Aluminized Steel	5000526	
Storm Collar	Stainless Steel/Galvanized	5000534	
Roof Flashing	0/12 – 6/12 6/12 - 12/12	5000542 5000550	
Horizontal Flashing Kit	Stainless Steel	5001037	
Wall Strap	Galvanized Steel	5000568	
Vertical Support Clamp	Galvanized Steel	5000576	
Firestop Support – Flat	Galvanized Steel	5000584	

* Special order only

VENTING PARTS LIST (CONT.)

Description	Color / Size	Part #	
6" Straight Pipe	Black	5000055	
	Blushtone*	5000063	
	Hunter Green*	5000089	
12" Straight Pipe	Black	5000170	
	Burgundy*	5000196	
	Hunter Green*	5000203	
24" Straight Pipe	Black	5000237	
	Blushtone*	5000245	
	Hunter Green*	5000261	
36" Straight Pipe	Black	5000295	
	Blushtone*	5000302	
	Hunter Green*	5000328	
48" Straight Pipe	Black	5000352	
	Blushtone*	5000360	
	Hunter Green*	5000386	
45 Degree Elbow	Black	5000419	
	Blushtone*	5000427	
	Hunter Green*	5000443	
90 Degree Elbow	Black	5000477	
	Blushtone*	5000485	
	Hunter Green*	5000500	
Firestop Support Cathedral Ceiling	Black	5000592	
	Blushtone*	5000609	
	Hunter Green*	5000625	
Trim Ring	Black	5000641	
	Blushtone*	5000659	
	Hunter Green*	5000675	
Starter Collar	Black	5000691	

* Special order only

VENTING PARTS LIST (CONT.)

Description	Color / Size	Part #	
Decorative Collar	Black	5000740	
	Blushtone*	5000758	
	Hunter Green*	5000774	
Sheet Metal Screw	Black	5000857	
	Blushtone	5000865	
	Hunter Green	5000881	

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